

For $\alpha, \beta: T \rightarrow K^\times$ irreps of T ,

$$\langle R(\alpha), R(\beta) \rangle_G = \langle \alpha, \beta \rangle_T + \langle \alpha, \beta^{-1} \rangle_T$$

We easily conclude

- If $\alpha^2 \neq 1$, then $R(\alpha)$ is irred, $\deg R(\alpha) = q+1$
- If $\alpha = 1$, then $R(1) = 1_G + St_G$ with 1_G the trivial rep/char of G , and St_G an irrep, called the Steinberg rep/char. $\deg(St_G) = q$.
- If p is odd and α_0 is the unique character of order 2, then $R(\alpha_0) = R_+(\alpha_0) + R_-(\alpha_0)$ with $R_\pm(\alpha_0)$ irreducible, nonisomorphic.
- If $\beta \neq \alpha^{\pm 1}$, then $\langle R(\alpha), R(\beta) \rangle = 0$

To find the degrees of $R_\pm(\alpha_0)$, we'll use $\tilde{G} = GL_2(\mathbb{F}_q)$
 $\tilde{T} = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in \tilde{G} \right\} \subseteq \tilde{B} = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \tilde{G} \right\}$

We still have $\tilde{G} = \tilde{B} \cup \tilde{B}s\tilde{B}$ and if we define

$$\tilde{R}(\tilde{\alpha}) = \text{Ind}_{\tilde{B}}^{\tilde{G}} \tilde{\alpha}_{\tilde{B}} \quad \text{for } \tilde{\alpha} \in K_0(K\tilde{T})$$

the same argument as for G gives

$$\langle \tilde{R}(\tilde{\alpha}), \tilde{R}(\tilde{\beta}) \rangle_{\tilde{G}} = \langle \tilde{\alpha}, \tilde{\beta} \rangle_{\tilde{T}} + \langle \tilde{\alpha}, {}^s\tilde{\beta} \rangle_{\tilde{T}}$$

$$\text{where } {}^s\tilde{\beta} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} = \tilde{\beta}(s^{-1} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} s) = \tilde{\beta} \begin{pmatrix} d & 0 \\ 0 & a \end{pmatrix}$$

So if $\tilde{\alpha}: \tilde{T} \rightarrow K^\times$ is a hom with ${}^s\tilde{\alpha} \neq \tilde{\alpha}$, then $\tilde{R}(\tilde{\alpha})$ is irreducible.

Lemma Assume p is odd. Then $\exists \tilde{\alpha}: \tilde{T} \rightarrow K^\times$ a hom such that $\tilde{\alpha}|_T = \alpha_0$ and any such satisfies ${}^s\tilde{\alpha} \neq \tilde{\alpha}$.

Proof Let $a \in \mathbb{F}_q^\times$ be a generator. So a^{-1} is also a generator.

So

$$\{ \text{homs } \tilde{\alpha}: \tilde{T} \rightarrow K^\times \} \xrightarrow{\sim} \mu_{q-1} \times \mu_{q-1} \subseteq K^\times \times K^\times$$

$$\tilde{\alpha} \longmapsto \left(\tilde{\alpha} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}, \tilde{\alpha} \begin{pmatrix} 1 & 0 \\ 0 & a^{-1} \end{pmatrix} \right)$$

$$\bigcup \left\{ \tilde{\alpha}: \tilde{T} \rightarrow K^\times \text{ hom} \right\} \xrightarrow{\sim} \mu_{q-1} \ni \zeta \xrightarrow{\uparrow} \mu_{q-1} \ni \zeta^{-1}$$

$$\text{Then if } \tilde{\alpha} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} = \zeta, \quad {}^s\tilde{\alpha} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} = \tilde{\alpha} \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} = -\zeta \neq \zeta \quad \square$$

Choosing $\tilde{\alpha}$ as in the lemma $\tilde{R}(\tilde{\alpha})$ is irreducible, and since $\tilde{G} = G \cdot \tilde{B}$, so Mackey's formula implies

$$\begin{aligned} \text{Res}_G^{\tilde{G}} \tilde{R}(\tilde{\alpha}) &= \text{Res}_G^{\tilde{G}} \text{Ind}_{\tilde{B}}^{\tilde{G}} \tilde{\alpha}_{\tilde{B}} \\ &= \text{Ind}_B^G \alpha_{0,B} = R_+(\alpha_0) + R_-(\alpha_0) \end{aligned}$$

$\tilde{G}/G$ permutes $R_+(\alpha_0)$ and $R_-(\alpha_0)$

$$\Rightarrow \dim R_+(\alpha_0) = \dim R_-(\alpha_0) = \frac{\dim \tilde{R}(\tilde{\alpha})}{2} = \frac{q+1}{2}$$

So when p is odd, the noncuspidal irreps of G are

- 1_G the trivial rep
- $R_\pm(\alpha_0)$, two $\frac{q+1}{2}$ -dim irreps of G
- St_G a q -dim irrep
- $\left(\frac{q-3}{2}\right)$ irreps $R(\alpha)$ with $\dim q+1$, corr to pairs $\{\alpha, \alpha^{-1}\}$ of $\alpha: T \rightarrow K^\times$ homs with $\alpha^2 \neq 1$.

Next: The cuspidal reps. But first...

§ A quick primer on étale cohomology

X is a topological space, R a commutative ring.

$$\rightsquigarrow H_{\text{sing}}^n(X, R) = \text{the singular cohomology of } X \text{ with coefficients in } R$$

$$= H^n(C^\bullet(X, R))$$

$$\text{where } C^n(X, R) = \{ f: \{ \text{cts } \sigma: \Delta^n \rightarrow X \} \rightarrow R \} \text{ and}$$

$$d: C^n(X, R) \rightarrow C^{n+1}(X, R)$$

$$\text{by } df(\sigma) = \sum_{n=0}^{n+1} (-1)^{n+1} f(\sigma|_{\text{th face of } \Delta^{n+1}})$$

Now if X a quasiproj var $\mathbb{C}$

$\rightarrow X(\mathbb{C})$ can be topologized as a subspace of $\mathbb{P}^n(\mathbb{C}) = \text{a complex manifold}$

$$\rightsquigarrow H_{\text{sing}}^n(X(\mathbb{C}), R)$$

Q Can we do this algebraically using only X ?

First, let's reinterpret $H_{\text{sing}}^n(X, R)$ for a top sp X .

Let $\text{Open}(X) = \text{category of open sets } U \subseteq X, \text{ whose morphisms are inclusions } V \hookrightarrow U$

Recall/Learn: For any category $\mathcal{C}$, a $\mathcal{C}$ -valued presheaf on X is a functor $F: \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$

And if $\mathcal{C}$ is a category with small limits (e.g. Set, Grp, Ab, $R\text{-Mod}$, ...) then F is a sheaf if $\forall U \in \text{Open}(X)$ and open covering $U = \bigcup_{i \in I} U_i$,

$$F(U) \xrightarrow{\text{res}_{U_i}} \prod_{i \in I} F(U_i) \xrightarrow[\text{res}_{U_i \cap U_j}]{\text{res}_{U_i, U_j}} \prod_{i,j} F(U_i \cap U_j)$$
 is an equalizer diagram

